

Cryptology Theory-4

1. Decrypt the following ciphertext using the Hill cipher

i) Cipher text : HLDTRZHXWQCMAG, Key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

Use (A=0, B=1, ..., Z=25). Find the plain text.

Soln: Inverse

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} = (3 \times 7) - (5 \times 2) = 21 - 10 = 11$$

$$11^{-1} \text{ mod } 26 \Rightarrow 26 = 11(2) + 4, 4 = 26 - 11(2)$$

$$11 = 4(2) + 3, 3 = 11 - 4(2)$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2))$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) - 11$$

$$1 = (-7)11 + 3(26), -7 + 26 = 19$$

Decryption: $P = K^{-1} C \text{ mod } 26$

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$\text{ci)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 7 + 14 \times 11 \\ 9 \times 7 + 5 \times 11 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$\text{cii)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 3 + 14 \times 19 \\ 9 \times 3 + 5 \times 19 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 15 \\ 18 \end{bmatrix}$$

$$\text{ciii)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 16 + 14 \times 8 \\ 9 \times 16 + 5 \times 8 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\text{civ)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 7 + 14 \times 9 \\ 9 \times 7 + 5 \times 9 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 17 \\ 4 \end{bmatrix}$$

$$(vi) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} = \begin{bmatrix} 3 \times 3 + 14 \times 23 \\ 9 \times 3 + 5 \times 23 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 19 \\ 12 \end{bmatrix}$$

$$(vii) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 3 \times 22 + 14 \times 16 \\ 9 \times 22 + 5 \times 16 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 290 \\ 278 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 4 \\ 18 \end{bmatrix}$$

$$(viii) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 3 \times 2 + 14 \times 12 \\ 9 \times 2 + 5 \times 12 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 18 \\ 0 \end{bmatrix}$$

$$(ix) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \times 0 + 14 \times 6 \\ 9 \times 0 + 5 \times 6 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

Plaintext: TOPSECRETMESSAGE

2. Consider the Hill cipher Key matrix

$K = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$, cipher text: RXAODY. Find the plain text for

the above cipher text.

Soln: Inverse:

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} = (24 \times 14) - (5 \times 19) = 241 \div 26 = 9 \text{ rem } 7 \Rightarrow 26 \times 9 =$$

$$234 - 241 = 7 \text{ mod } 26$$

$$7^{-1} \text{ mod } 26 \Rightarrow 26 = 7(3) + 5, 5 = 26 - 7(3)$$

$$7 = 5(1) + 2, 2 = 7 - 5(1)$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = (-11)7 + 3(26) \Rightarrow -11 + 26 = 15$$

$$15 \begin{bmatrix} 14 & 19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & -285 \\ -75 & 360 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 2 & 5 \\ 3 & 21 \end{bmatrix}$$

$$(i) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 \times 17 + 1 \times 23 \\ 3 \times 17 + 22 \times 23 \end{bmatrix} = \begin{bmatrix} 57 \\ 557 \end{bmatrix} \pmod{26} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{bmatrix} F \\ L \end{bmatrix}$$

$$(ii) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 \times 0 + 1 \times 14 \\ 3 \times 0 + 22 \times 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \pmod{26} = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{bmatrix} O \\ W \end{bmatrix}$$

$$(iii) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 \times 3 + 1 \times 24 \\ 3 \times 3 + 22 \times 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{bmatrix} E \\ R \end{bmatrix}$$

plaintext : FLOWER.

3) Confusion vs. Diffusion

Confusion: It hides the relationship between the plaintext & ciphertext by changing the value of characters.

Diffusion: It spreads the information of the plaintext by changing the position of characters.

Cipher	Changes character value?	Changes character position?	Confusion / Diffusion	Justification
Shift Cipher	Yes	No	Confusion	It shifts each letter by a fixed number.
Multiplicative Cipher	Yes	No	Confusion	It multiplies the numerical value of each letter by Key.
Affine Cipher	Yes	No	Confusion	It changes each letter using formula $C = aP + b \pmod{26}$
Transposition Cipher	No	Yes	Diffusion	It keeps the same letters but changes their positions.

* Relationship between substitution & confusion

- Substitution changes one character into another character, Therefore, substitution mainly provides confusion.

Example:

$A \rightarrow D, B \rightarrow E, C \rightarrow F$

The character values are changed, but their positions remain same.

* Relationship between transposition and diffusion

- Transposition does not change the characters. It only changes their positions. Therefore, transposition mainly provides diffusion.

Example:

Plaintext: HELLO

After transposition: LHOEL

The letters are the same, but their positions are different

9) Transposition Encryption

Encrypt the message "The arrow is pointing north direction" using.

i) Key = 4

T H E A
R R O W
I S P O
I N T I
N G N O
R T H D
I R E C
T I O N

Ciphertext:

TRIINRITHRSNRTIEOPTNHEDAWOIODEN

ii) Key = ATLAS

ATLAS
1 5 3 2 4

A T L A S
1 5 3 2 4
T H E A R
R O W I S
P O I N T
I N G N O
R T H D I
R E C I I
O N X X X

Ciphertext: TRPIRROAINNOTIXENI4HLCXRSTOITIXHOONTEN

(iii) Double
 Key 1 : BREAK
 Key 2 : Log

BREAK
 25413

B R E A K
 2 5 4 1 3
 T H E A R
 R O W I S
 P O I N T
 I N G N O
 R T H O I
 R E C T I
 O N X X X

ciphertext : AINNDTXTRPIRRORSTOIXEWIGHCX HOONTEN

Log
 231

L O 4
 2 3 1
 T H E
 A R R
 O W I
 S P O
 I N T
 I N 4
 N O R
 T H D
 I R E
 C T I
 O N X

ciphertext : ERIOTGRDIX TAOSIINTICO HRWPNNOHRTN

5) Transposition Decryption

(i) Key 10

Ciphertext : TIDIE MRNME REO40 SANOW RSLTP EEEDO SSSNU
 AHTUD BIENP GODAE PEIXD BLNSX = 60/10 = 6

6
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T	R	O	O	P	S	H	E	A	D
I	N	4	W	E	S	T	N	E	E
D	M	O	R	E	S	U	P	P	L
I	E	S	S	E	N	D	4	E	N
E	R	A	L	D	U	B	O	I	S
M	E	N	T	O	A	I	D	X	X

Plaintext : TROOPS HEADING WEST NEED MORE SUPPLIES
 SEND GENERAL DUBOIS MEN TO AID

(ii) Keyword : SECRET

Ciphertext: ILEDTHPTRSOIITMAXSAXERXTANONISNFOTX

$$36 \div 6 = 6$$

S	E	C	R	E	T
5	2	1	4	3	6
T	H	I	S	I	S
A	P	L	A	I	N
N	T	E	X	T	F
O	R	D	E	M	O
N	S	T	R	A	T
I	O	N	X	X	X

C1:	ILEDTH
C2:	TPTRSO
C3:	ITMAX
C4:	SAXERX
C5:	TANONI
C6:	SNFOTX

plaintext: THIS IS A PLAIN TEXT FOR DEMONSTRATION

(iii) Keyword: PLAIN

Ciphertext: VEY EAX ARA ATX HQD = 15 ÷ 5 = 3

P	L	A	I	N
5	3	1	2	4
H	A	✓	E	A
4	R	E	A	T
D	A	Y	X	X

C1:	VEY
C2:	EAX
C3:	ARA
C4:	ATX
C5:	HQD

plaintext: HAVE A GREAT DAY

6) Permutation

Find the inverse permutation for

1. 4 3 1 5 2

original
position

1	2	3	4	5
4	3	1	5	2

$$p^{-1} = 3 5 2 1 4$$

$$2) \quad 7413526$$

original position: 1 2 3 4 5 6 7
 7 4 1 3 5 2 6

$$P^{-1} = 3 \ 6 \ 4 \ 2 \ 5 \ 7 \ 1$$

$$3) \quad 4132$$

original position: 1 2 3 4
 4 1 3 2

$$P^{-1} = 2 \ 4 \ 3 \ 1$$

$$4) \quad 35142$$

original position: 1 2 3 4 5
 3 5 1 4 2

$$P^{-1} = 3 \ 5 \ 1 \ 4 \ 2$$